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DC27 EMBA
Assignment 2 Stats

1. See table

20	0.05	1
↓		↓
30	0.05	1.5

$\mu = 25.41$
Variance = 7.50
stdev = 2.74

2.

a. within on stdev
25.41 ± 2.74

$22.67 \leq x \leq 28.15$ ✓ Q2a
(23, 24, 25, 26, 27, 28)
$0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.1 = 0.67$

b. within two stdev
$25.41 \pm 2 \cdot (2.74)$

$19.9 \leq x \leq 30.9$ ✓ Q2b
Every number $P = 100\% = 1.00$

3.

		Value	Exp Value	
20	0.05	Ø		-1.36
21	0.06	Ø	[P = 0.47]	-1.36
22	0.07	Ø	Ø	-1.36
23	0.08	Ø		-1.36
24	0.09	Ø		-1.36
→ 25	0.12	Ø		-1.36
26	0.15	1	0.15	-.036
27	0.13	2	0.26	[P = 0.53] Positive value
28	0.10	3	0.3	
29	0.10	4	0.4	
30	0.05	5	0.25	

Expected prod is $1.36

A

4. The probability of a positive value is 0.53

5. $\mu = \$1.36$

for std dev, see table

variance 2.57
stdev 1.60